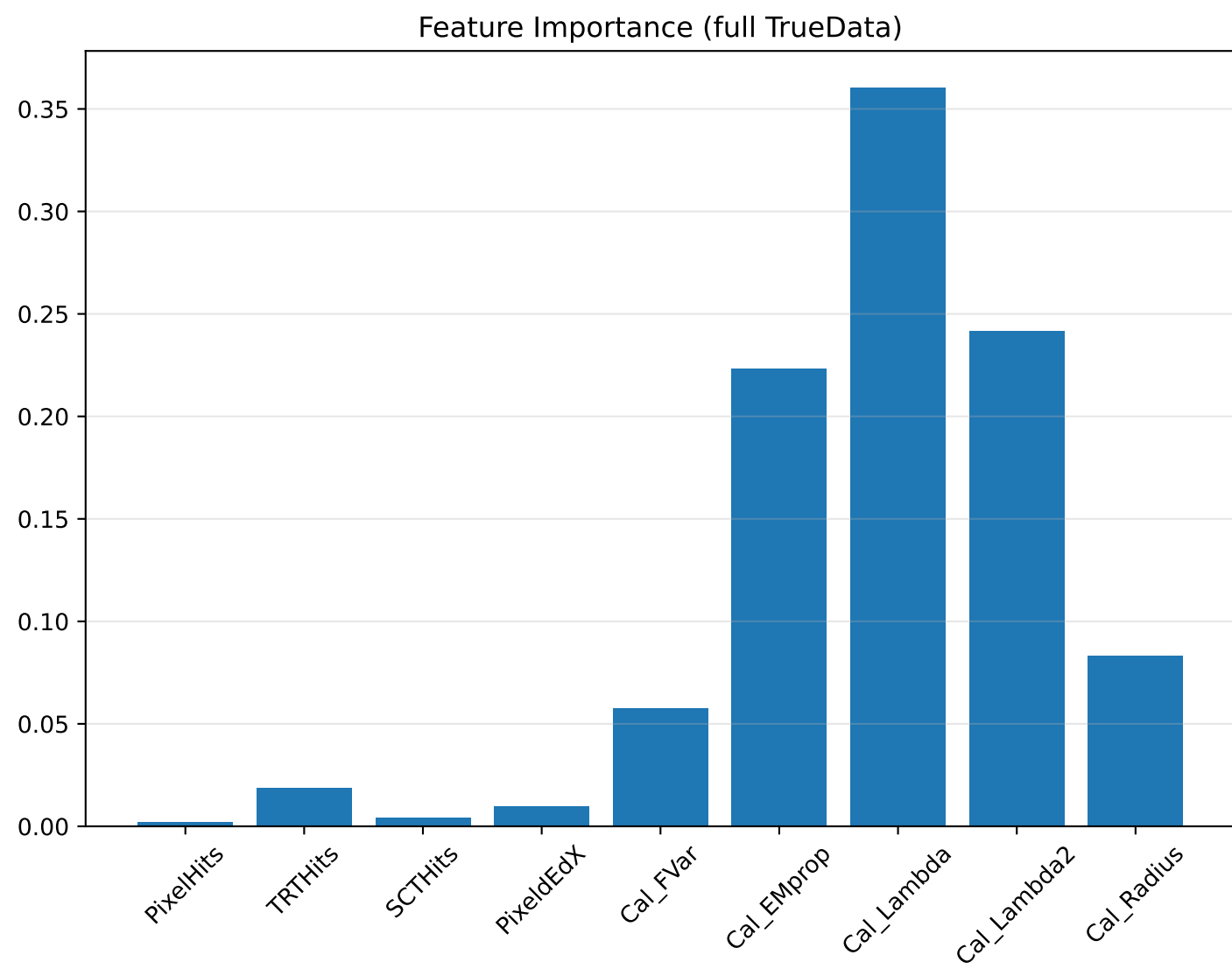
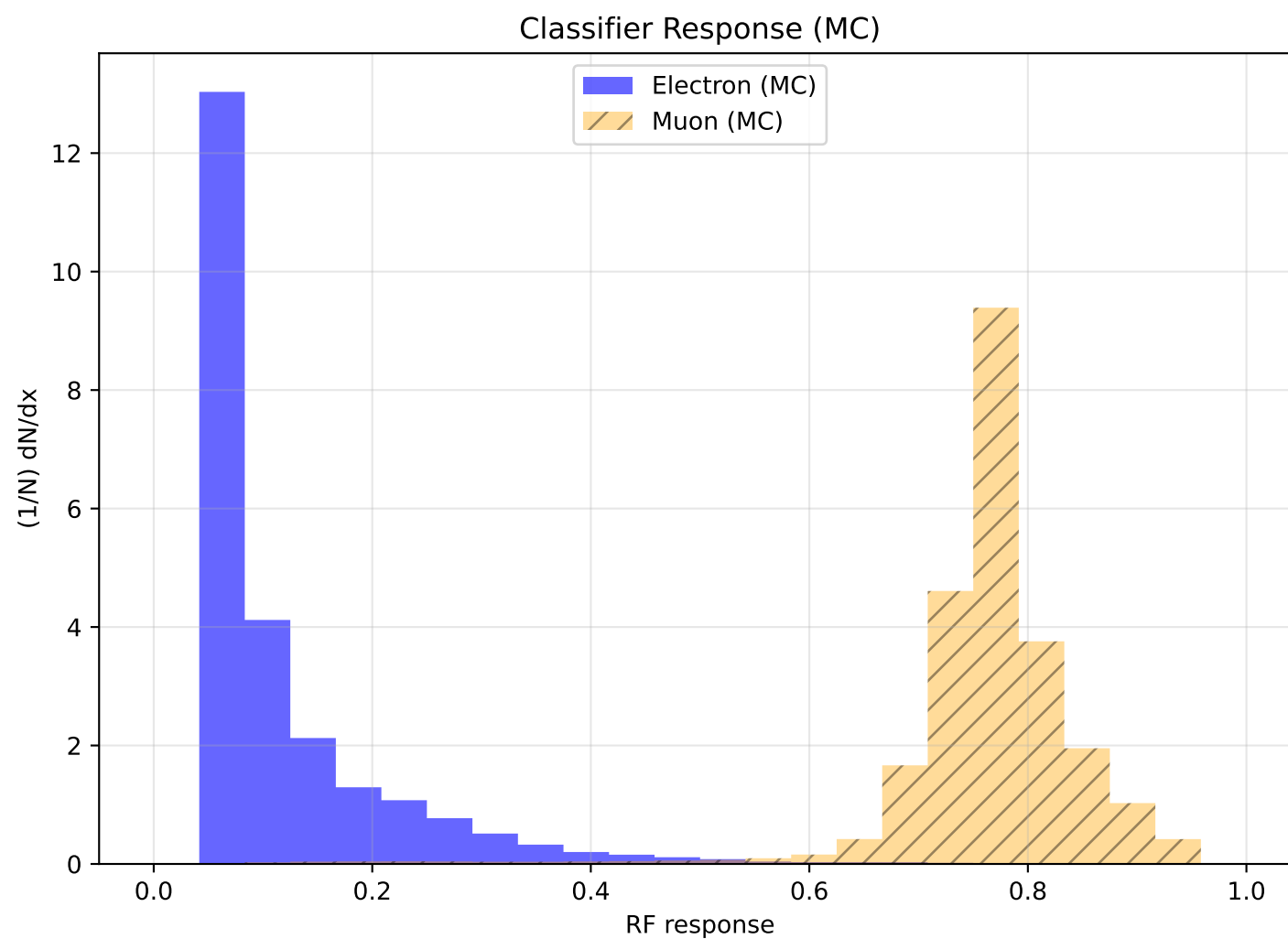
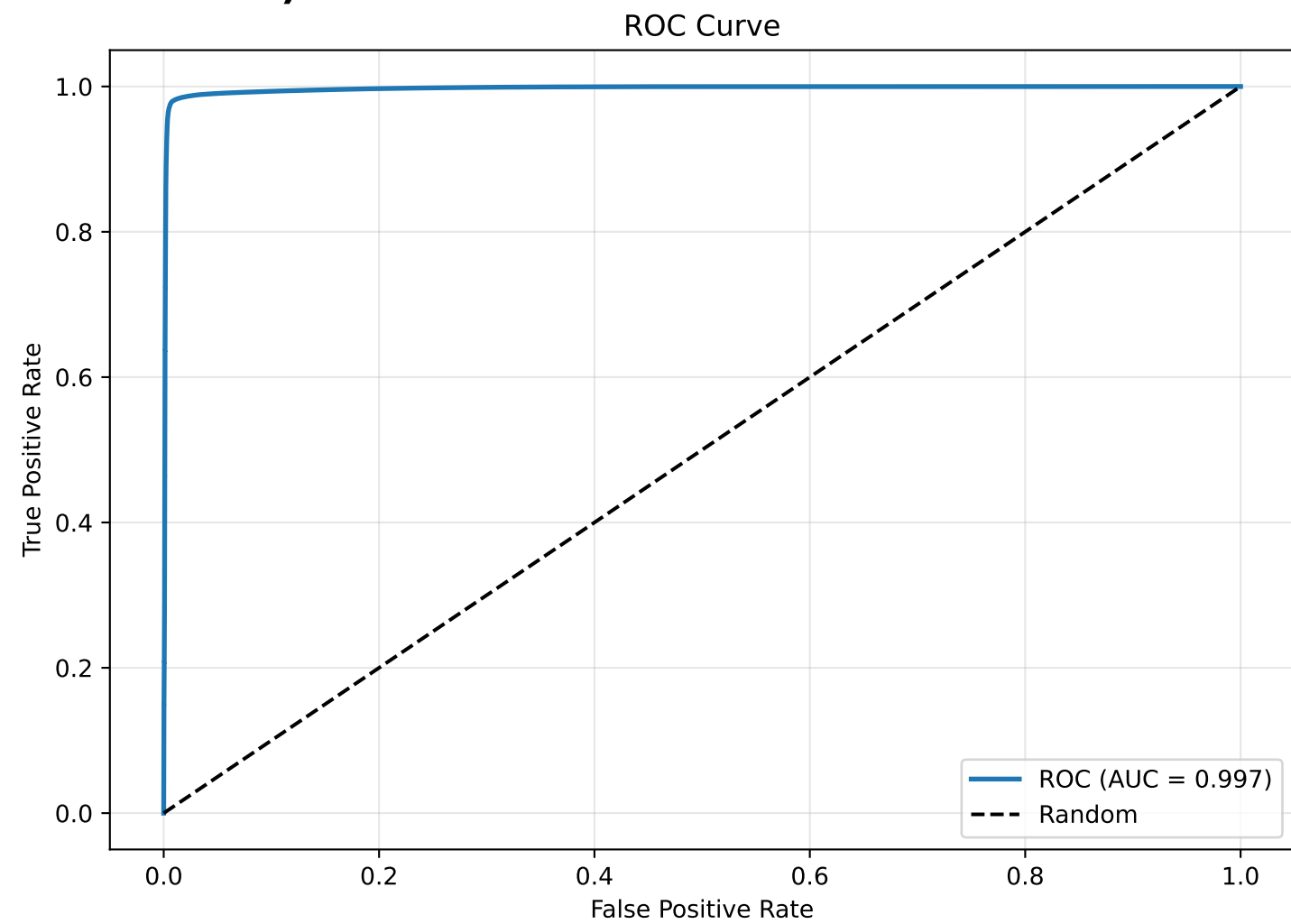
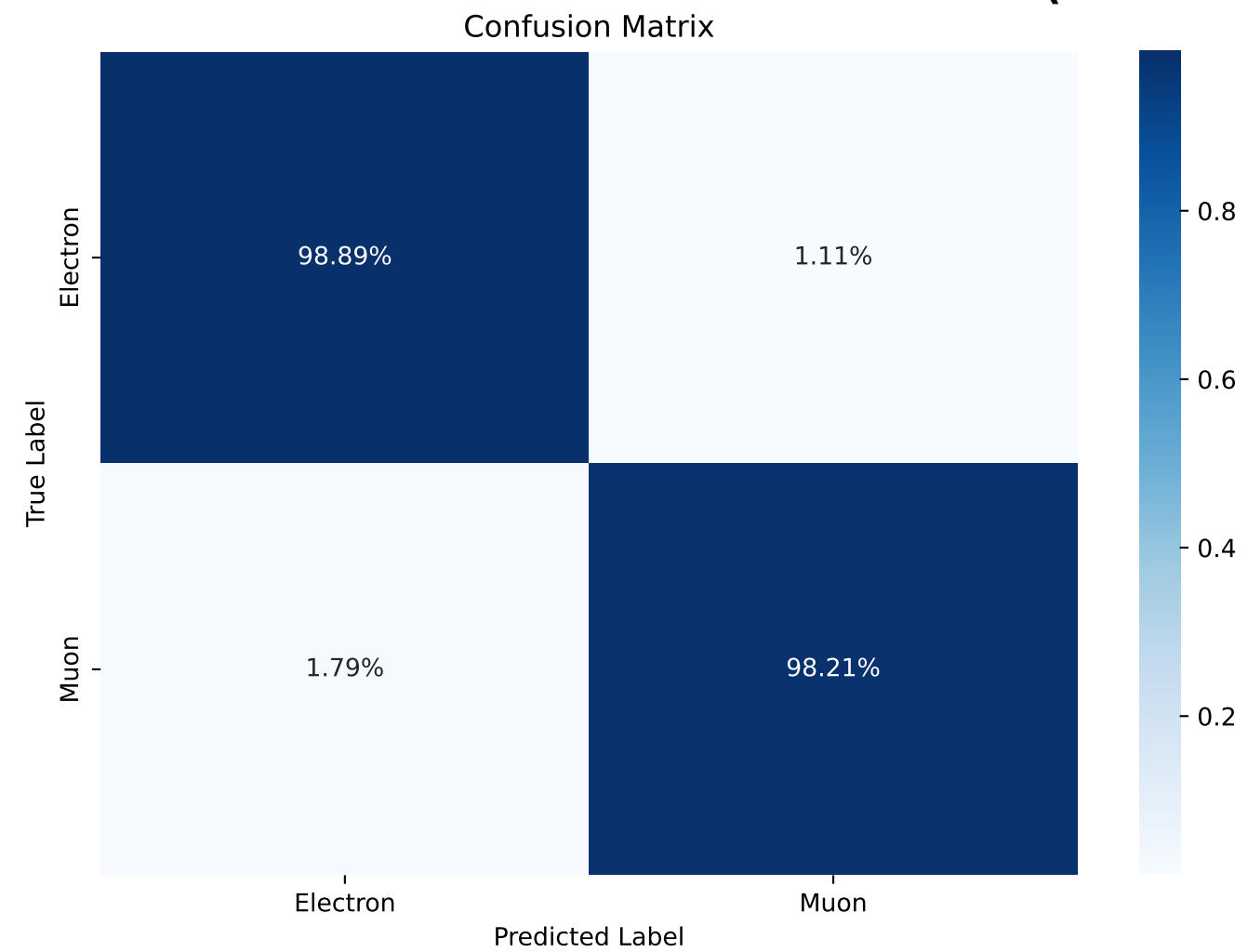
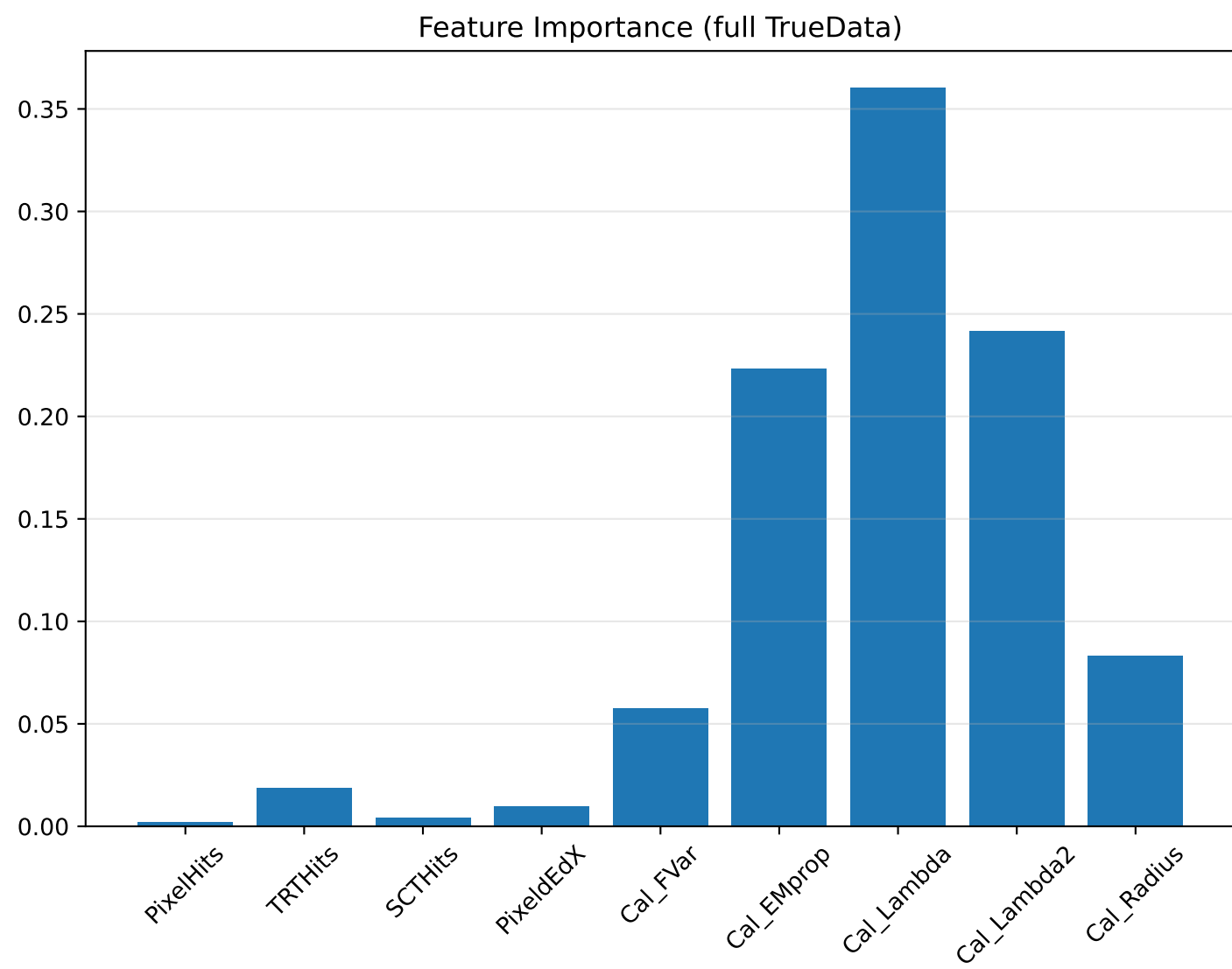
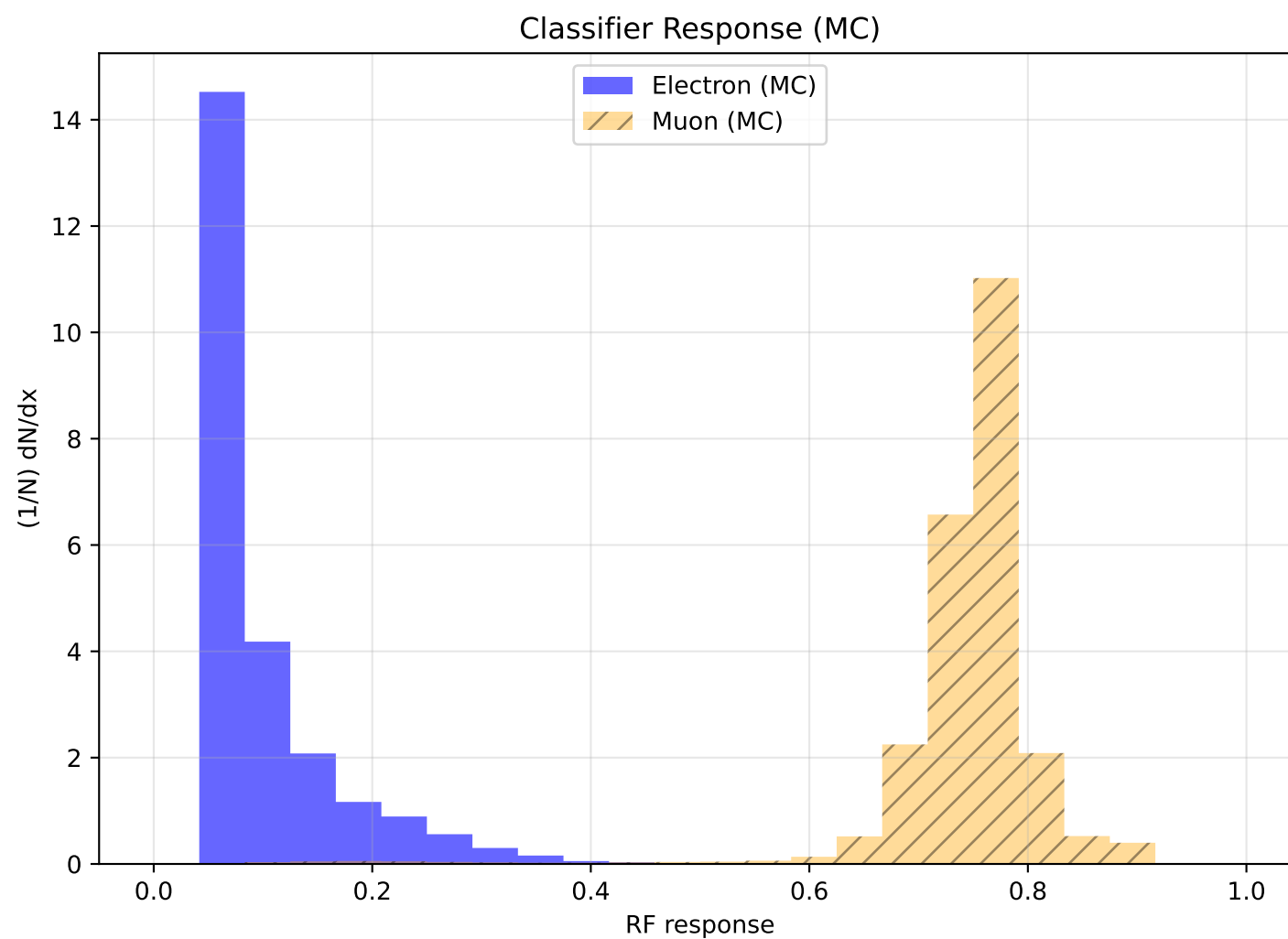
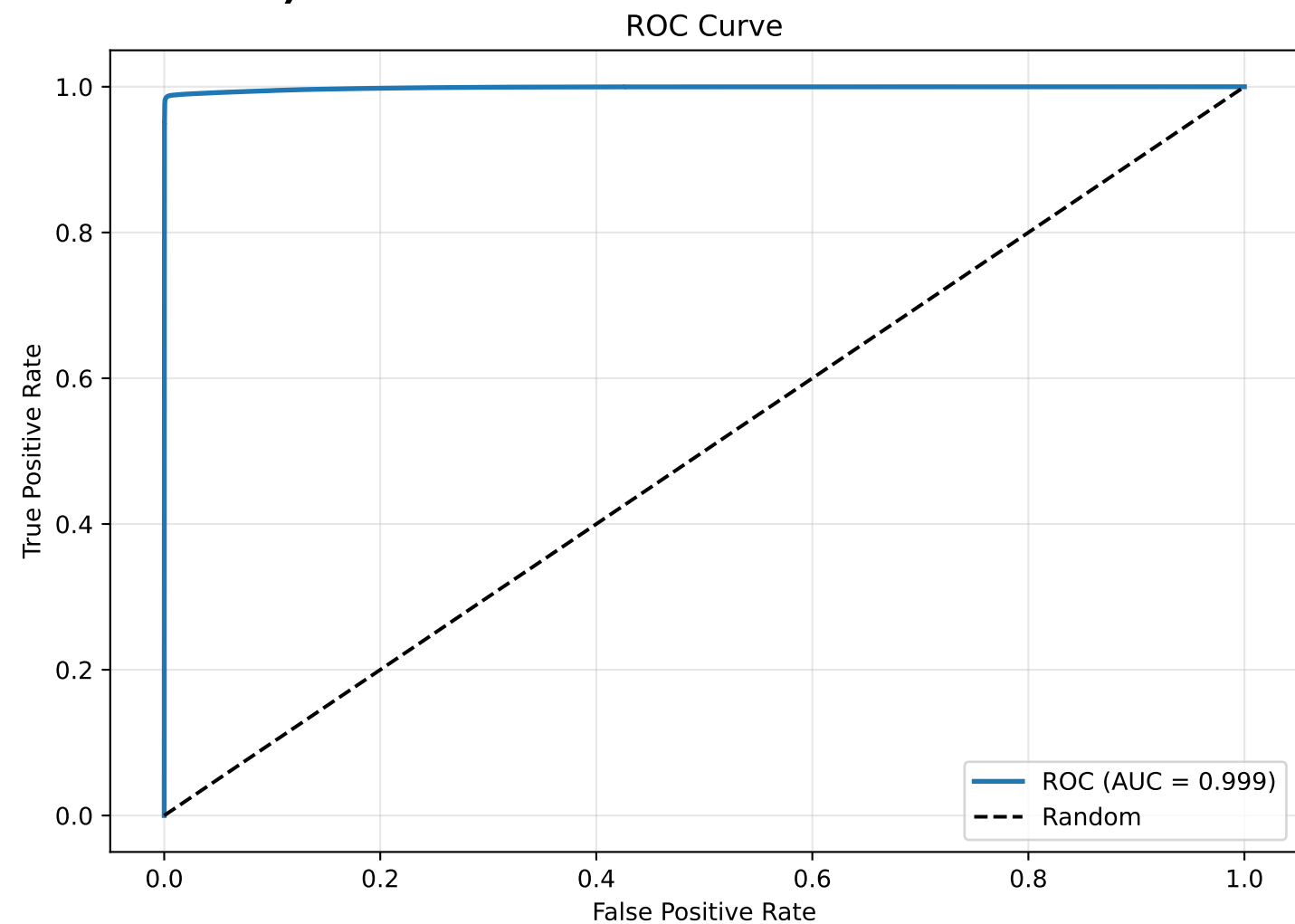
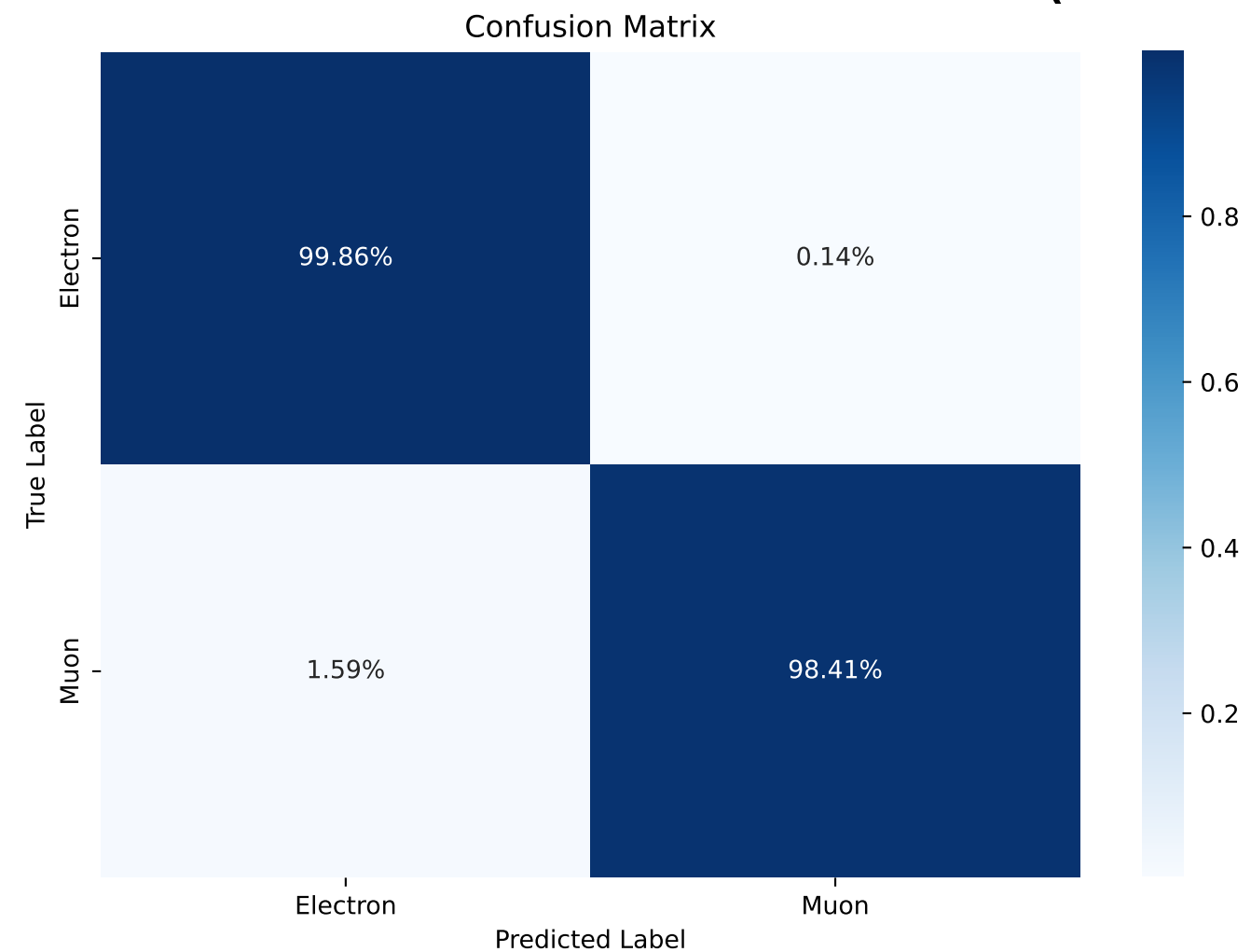


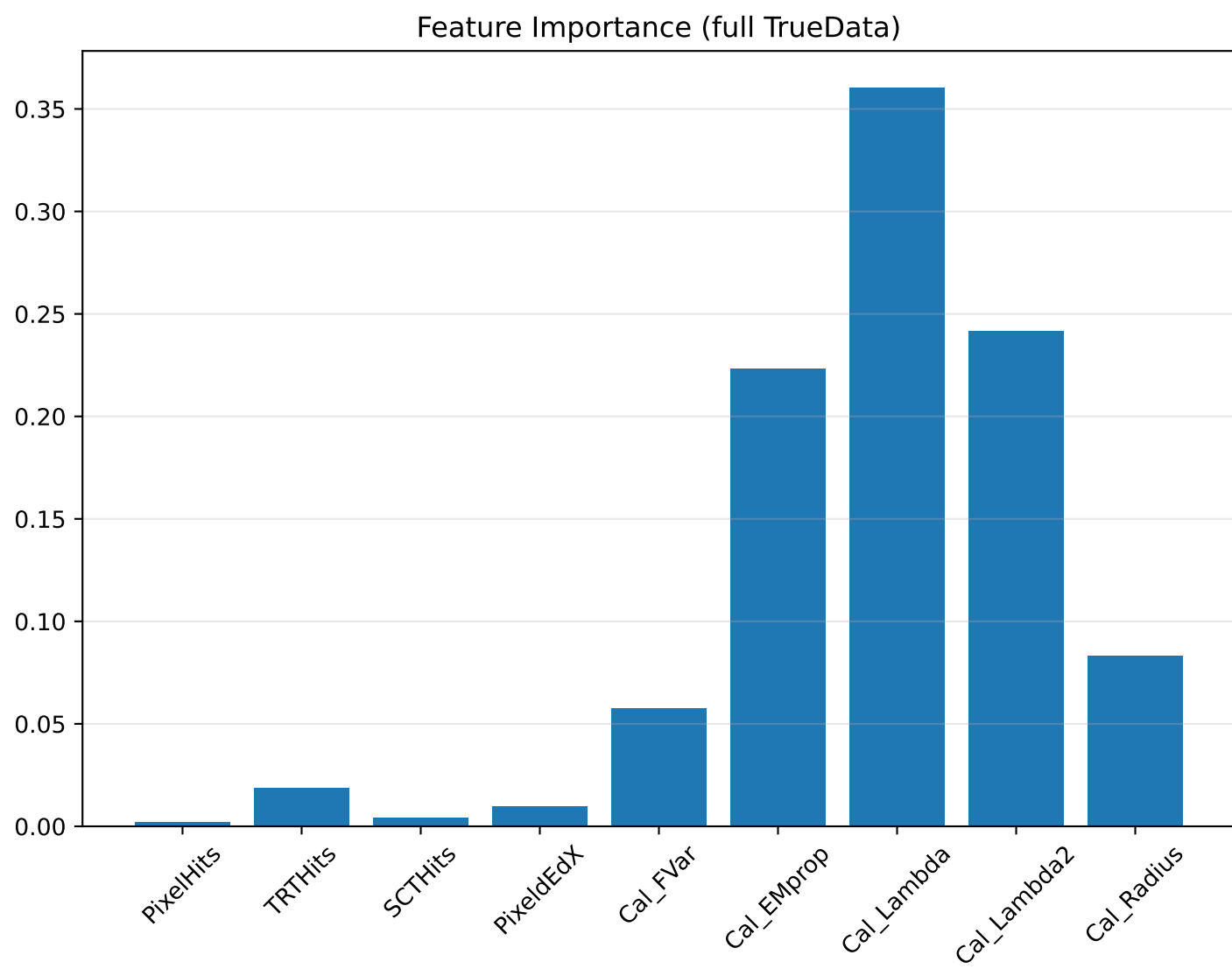
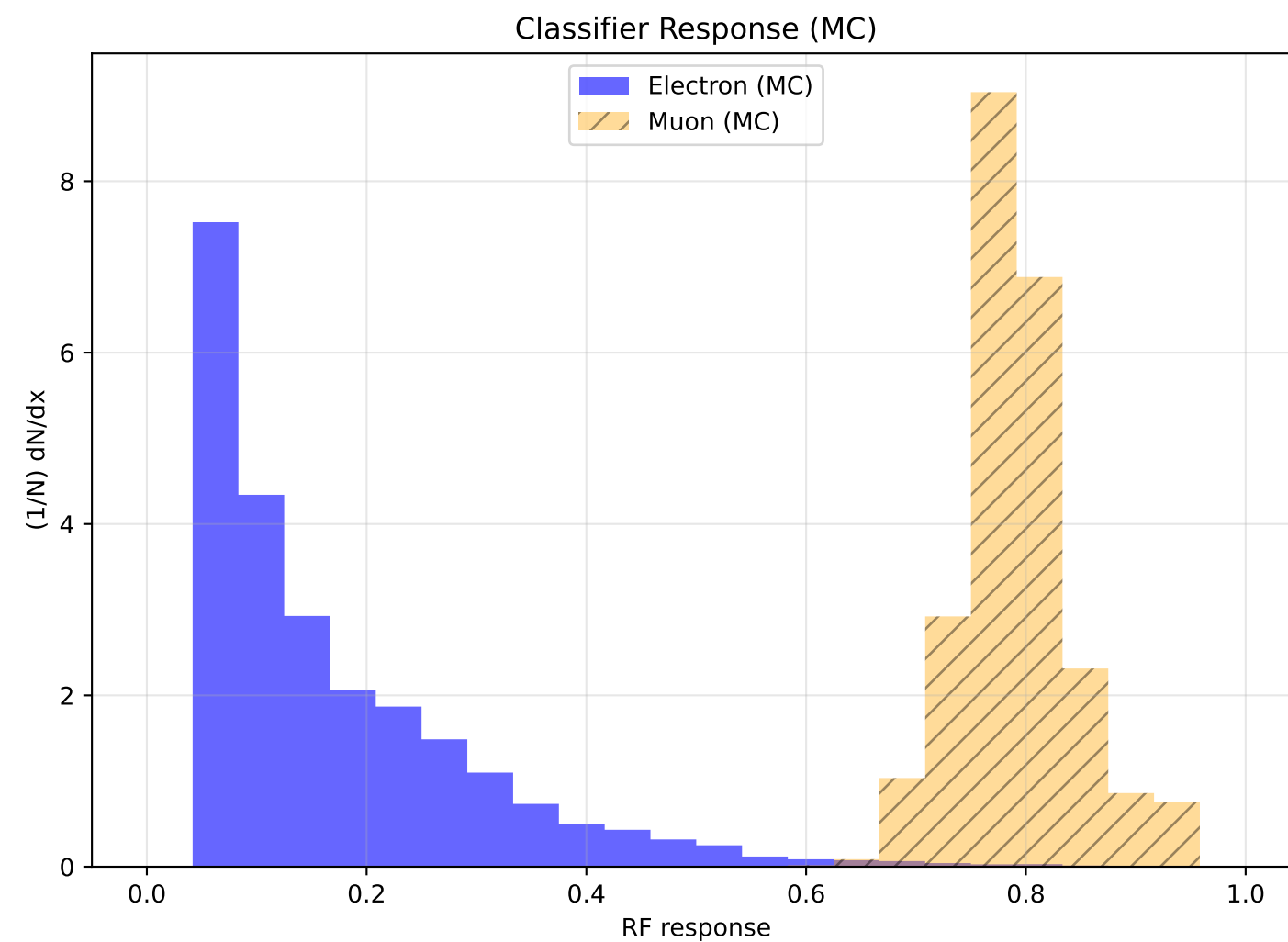
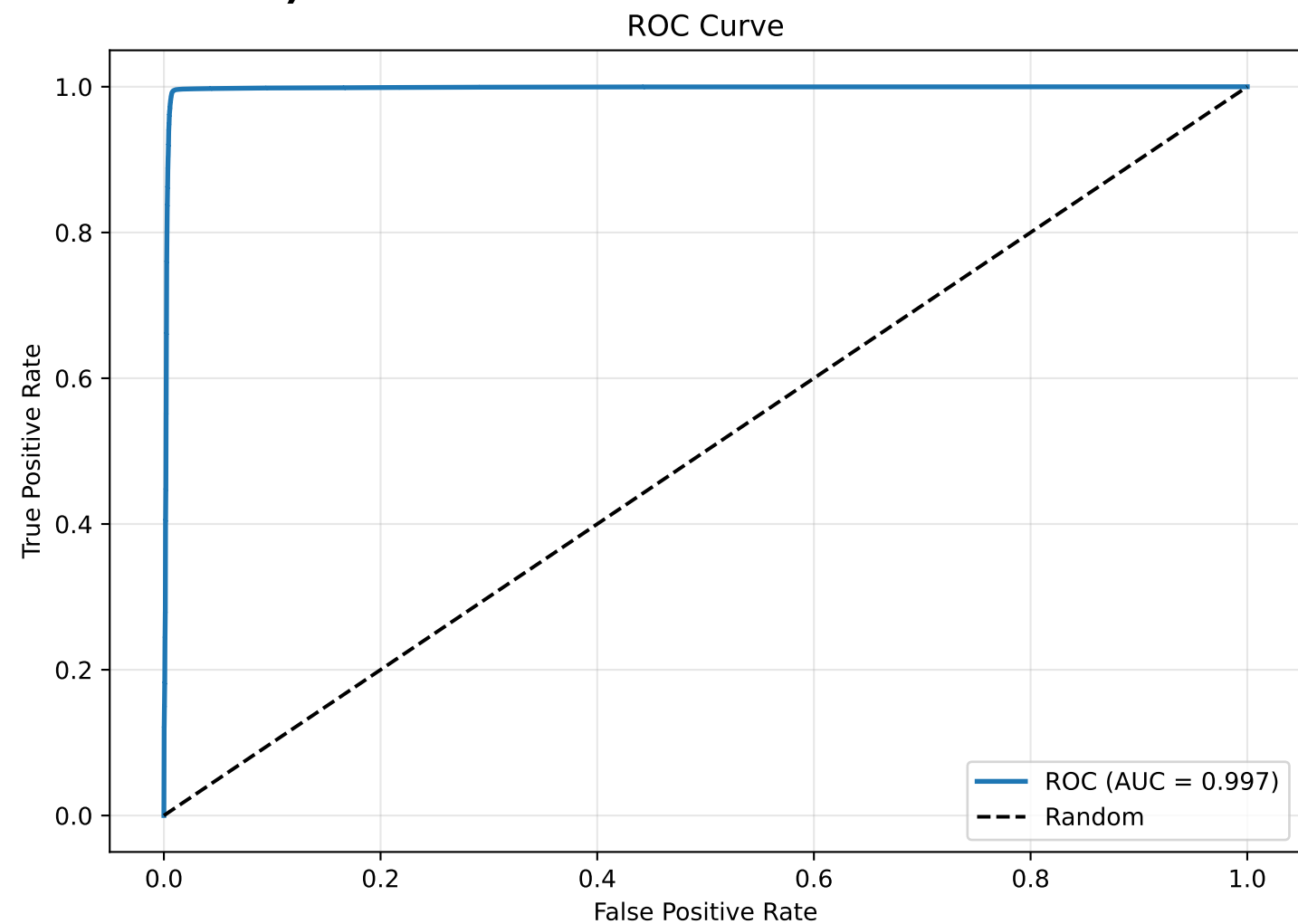
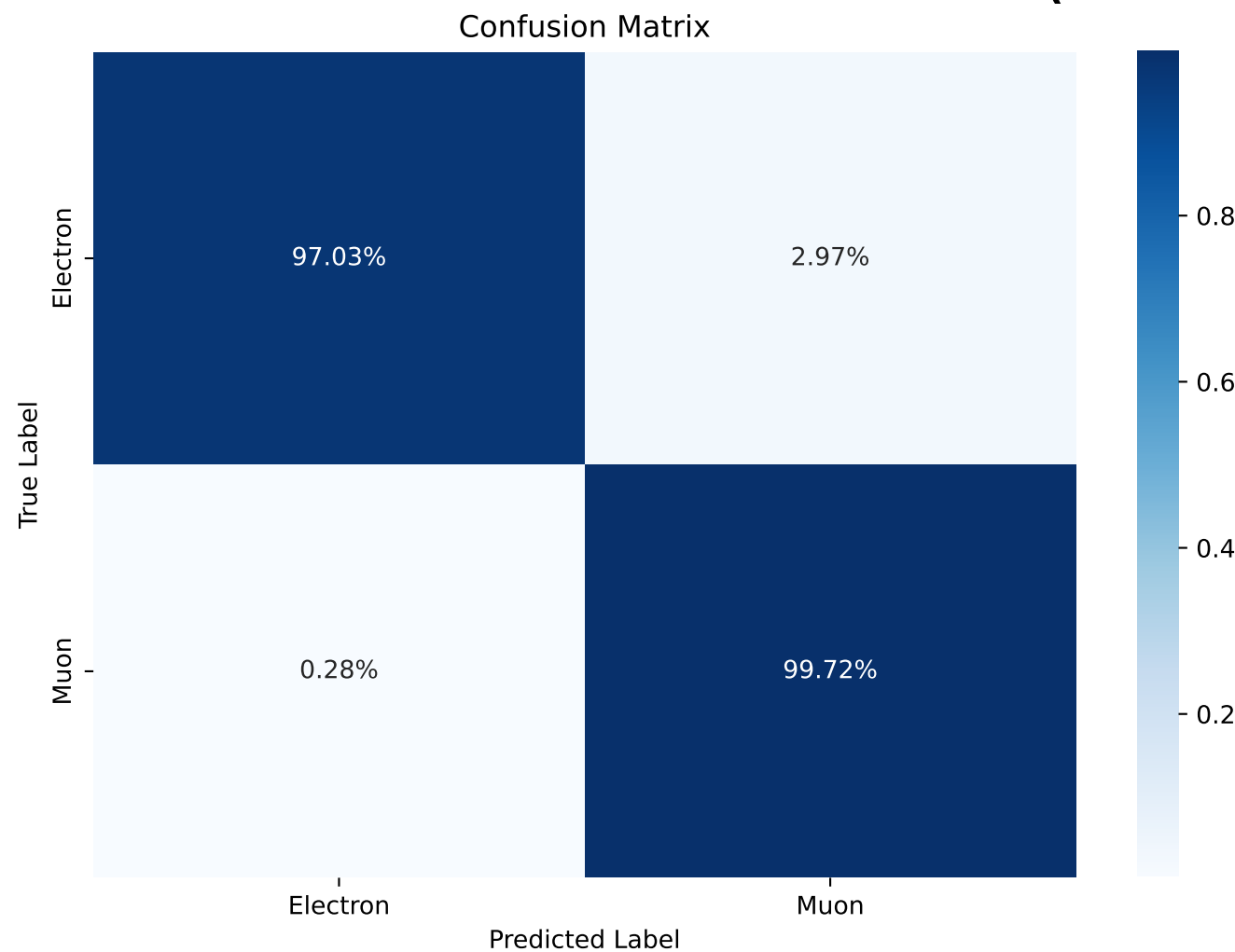
**EtaRange = FullRange - Accuracy on MC: 0.983  
(trained on full TrueData)**



**EtaRange = <1.0 - Accuracy on MC: 0.986  
(trained on full TrueData)**

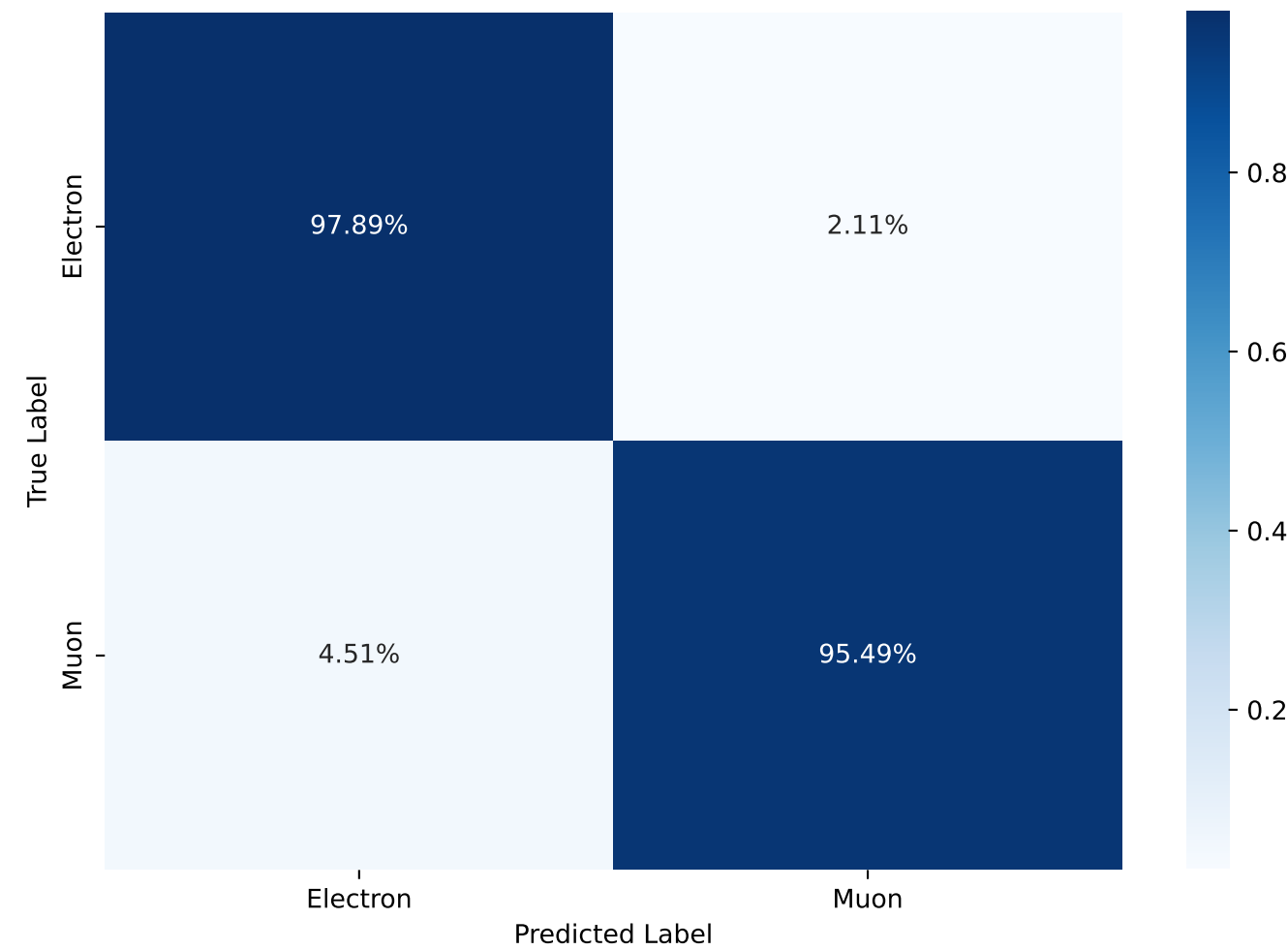


**EtaRange = 1.0-1.5 - Accuracy on MC: 0.995  
(trained on full TrueData)**

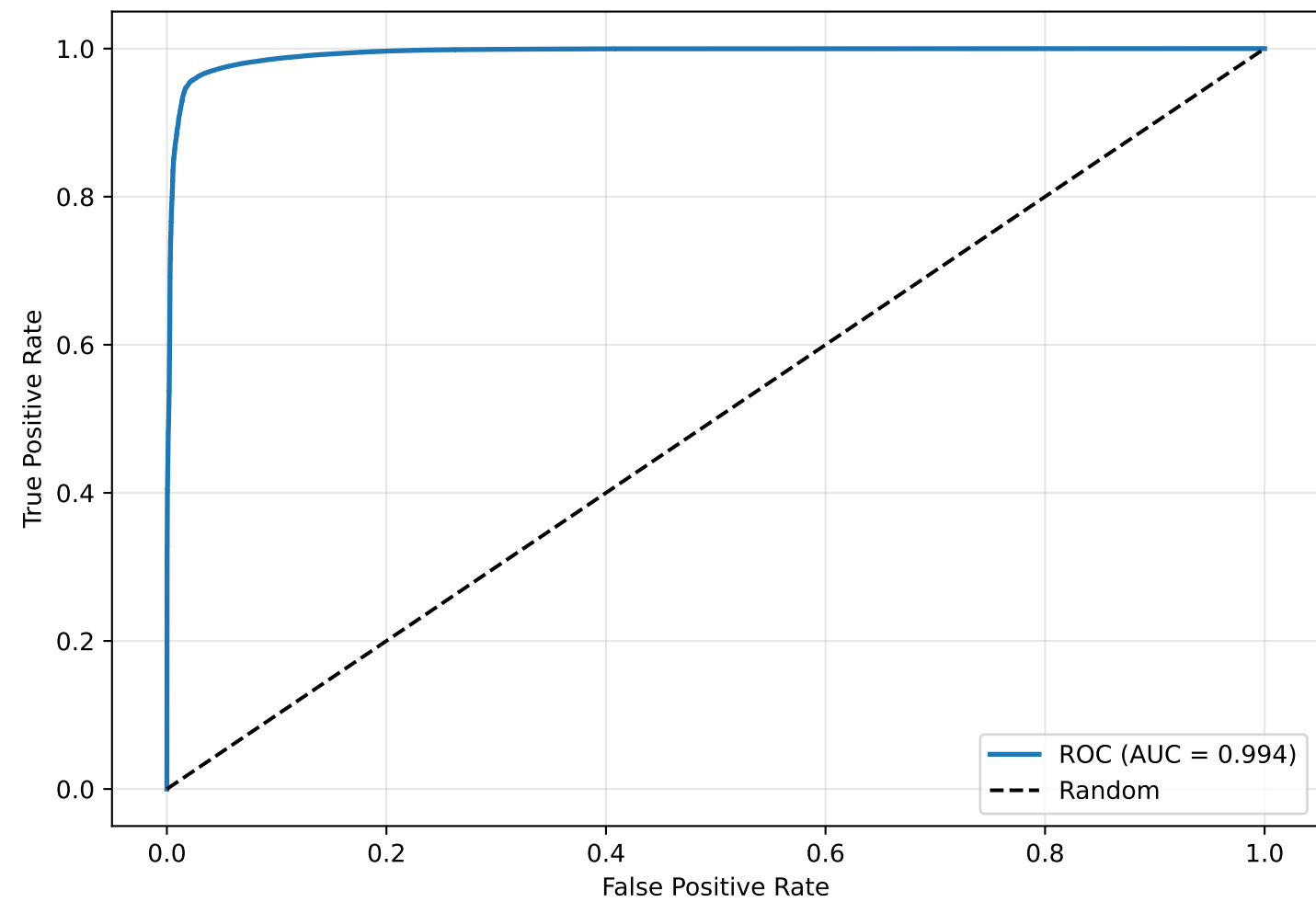


# EtaRange = 1.5-2.5 - Accuracy on MC: 0.958 (trained on full TrueData)

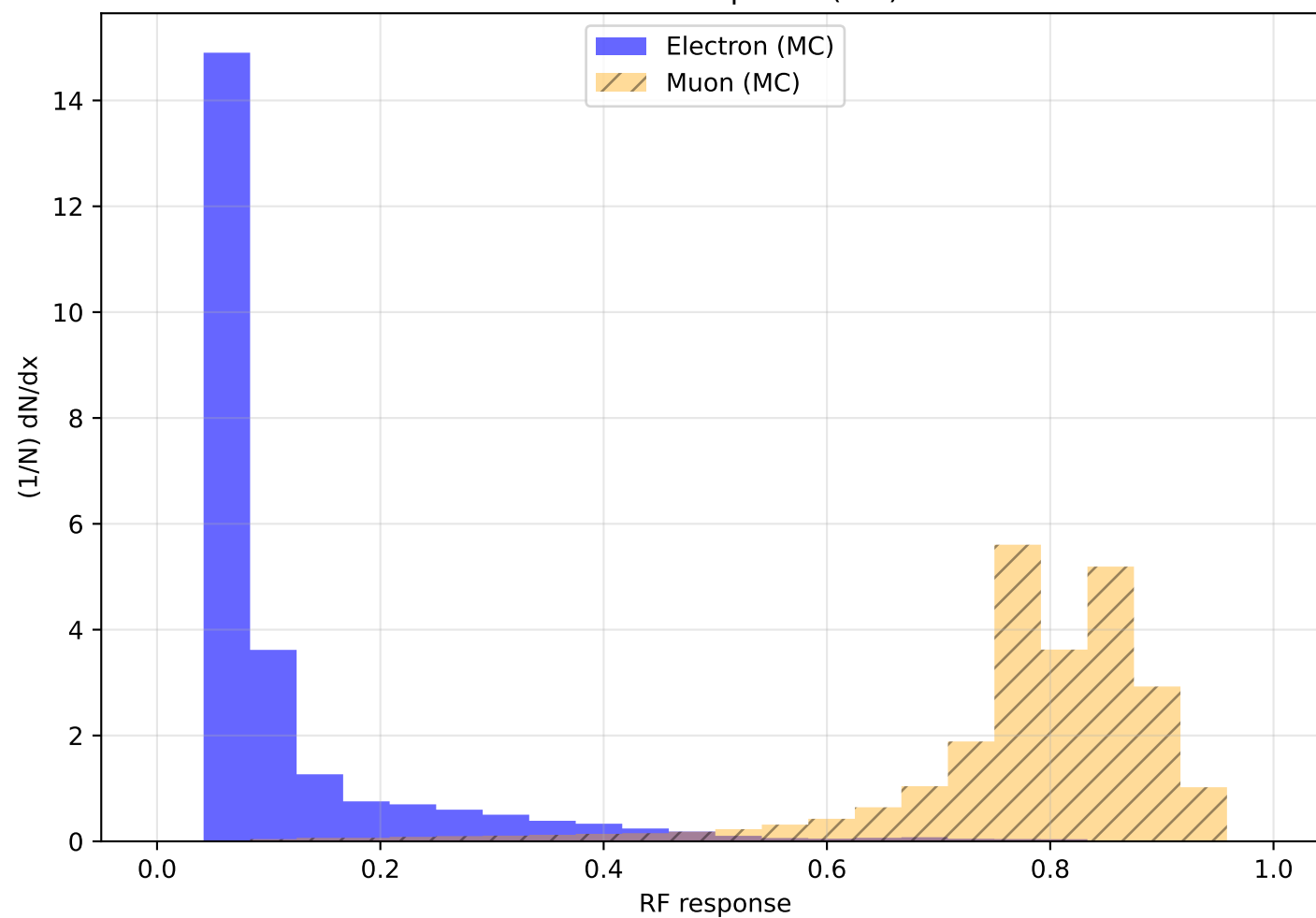
Confusion Matrix



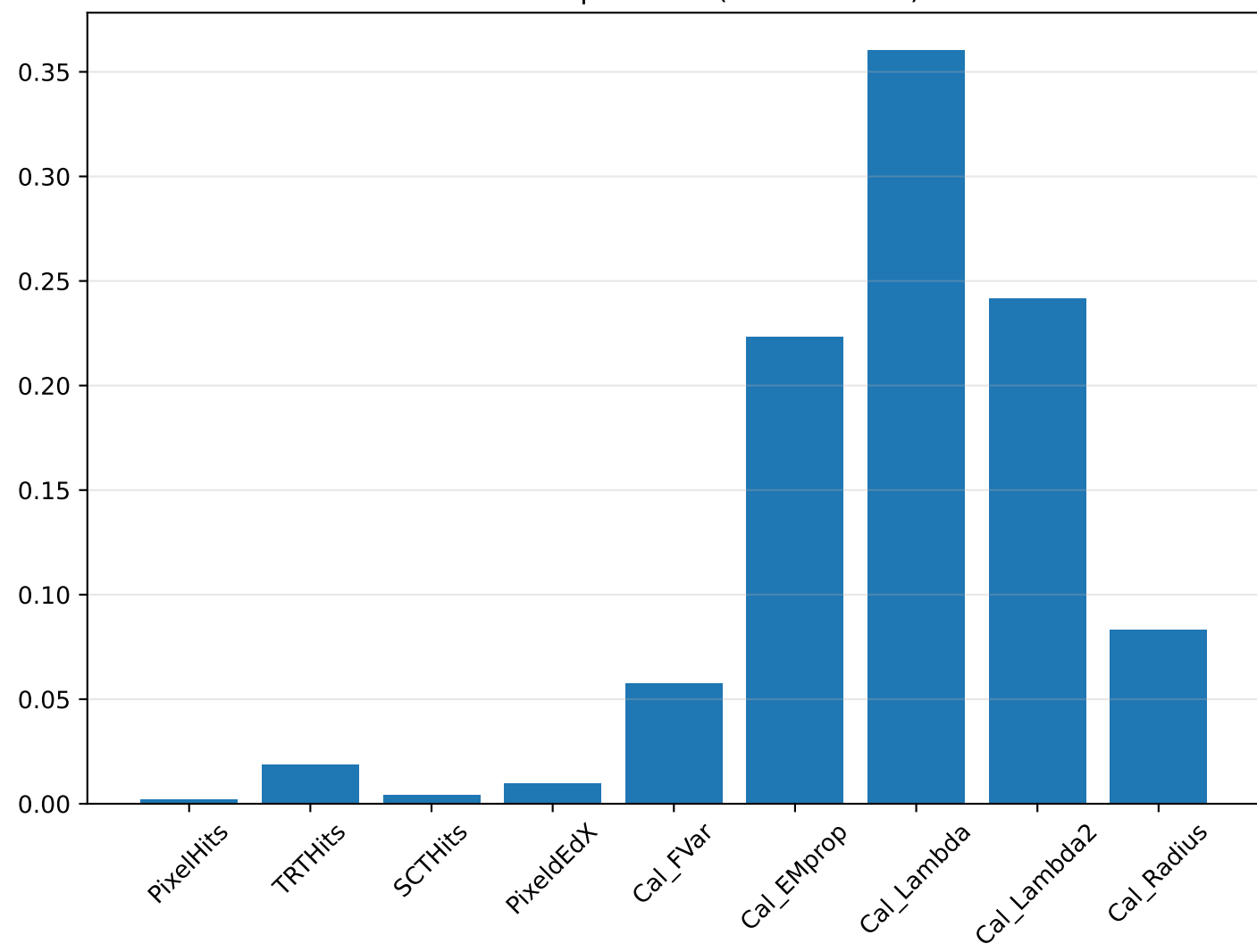
ROC Curve



Classifier Response (MC)



Feature Importance (full TrueData)



# Classification Reports Summary

## Trained: Mixed TrueData.root | Test: MC Samples

### EtaRange = FullRange

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Electron     | 0.89      | 0.99   | 0.93     | 154464  |
| Muon         | 1.00      | 0.98   | 0.99     | 1098142 |
| accuracy     |           |        | 0.98     | 1252606 |
| macro avg    | 0.94      | 0.99   | 0.96     | 1252606 |
| weighted avg | 0.98      | 0.98   | 0.98     | 1252606 |

### EtaRange = <1.0

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Electron     | 0.91      | 1.00   | 0.95     | 93264   |
| Muon         | 1.00      | 0.98   | 0.99     | 568558  |
| accuracy     |           |        | 0.99     | 661822  |
| macro avg    | 0.96      | 0.99   | 0.97     | 661822  |
| weighted avg | 0.99      | 0.99   | 0.99     | 661822  |

### EtaRange = 1.0-1.5

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Electron     | 0.97      | 0.97   | 0.97     | 34327   |
| Muon         | 1.00      | 1.00   | 1.00     | 313583  |
| accuracy     |           |        | 0.99     | 347910  |
| macro avg    | 0.99      | 0.98   | 0.98     | 347910  |
| weighted avg | 0.99      | 0.99   | 0.99     | 347910  |

### EtaRange = 1.5-2.5

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Electron     | 0.73      | 0.98   | 0.84     | 26873   |
| Muon         | 1.00      | 0.95   | 0.98     | 216001  |
| accuracy     |           |        | 0.96     | 242874  |
| macro avg    | 0.86      | 0.97   | 0.91     | 242874  |
| weighted avg | 0.97      | 0.96   | 0.96     | 242874  |